Shiyi Chen

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Education

University of California, Berkeley Aug 2025 – **Expected May 2029**
Bachelor's Degree in Computer Science | GPA: 4.0/4.0, Dean's List

- **Systems & software:** Data Structures, Efficient Algorithms, Computer Architecture, Digital Infrastructure
- **Math & AI:** Artificial Intelligence, Machine Learning, Reinforcement Learning, Discrete Mathematics, Probability Theory, Linear Algebra, Calculus

Technical Skills

- **Languages:** Python (primary), C, C++, RISC-V Assembly, Java, TypeScript, Dart
- **Systems & Tools:** Linux, Slurm, tmux, Git, CUDA, SQL, FastAPI, Flutter
- **ML & Data:** PyTorch, scikit-learn, HuggingFace, LangGraph, OpenCV, NumPy, Pandas, Matplotlib, SpaCy, NLTK

Research & Project Experience

Berkeley Artificial Intelligence Research (BAIR) May 2026 – Present
Undergraduate Researcher, advised by Yun S. Song

- Researching efficient approximate guidance for discrete flow matching models that generate protein sequences, steering sampling toward high-reward outputs at a fraction of exact guidance's cost.
- Hands-on with Gaussian Processes (probabilistic prediction with uncertainty estimates) and encoder-decoder transformers (encoder representations feeding a sequence-generating decoder).
- Implemented and benchmarked Exact Guidance and Bootstrap Sequential Monte Carlo on sinusoidal-wave and protein-sequence generative models; ran all experiments on a Linux GPU cluster with NVIDIA A100 GPUs using Slurm.

UC Berkeley & Lawrence Berkeley National Laboratory (LBNL) Jan 2026 – May 2026
Student AI Researcher

- Built multi-agent architectures to label bias in 3D point-cloud and raster data, reducing per-tree labeling time by **98%**.
- Designed a decoupling framework that compiles expert rules into executable decision trees, so the orchestrator agent can be reconfigured with zero code changes; outperformed ML baselines by **30 Macro-F1 points**.
- Co-authored *TreeAgent*, accepted to the ICML 2026 Workshop on AI for Science (top 50% among accepted submissions).

Jacobs Institute for Design Innovation Sep 2025 – May 2026
Independent ML Researcher / Developer

- Built a full-stack, cross-platform posture-detection and improvement app with Flutter, FastAPI, and PostgreSQL, with 80+ waitlist users and 2+ commercial collaborators.
- Selected from 200+ applicants as one of three undergraduate teams in the Berkeley Student Entrepreneurship Program (StEP), Fall 2025.
- Derived a training objective that maps mesh-based Human NeRF output to a fixed dimension for deep-model training [poster].

Systems Projects, UC Berkeley CS 61C Fall 2026
C and RISC-V

- **Snake game in C**: implemented the game state, update logic, and a memory-efficient board loader (`fgets`/`realloc`, no over-allocation for variable-length rows) using structs, pointers, and function pointers; wrote custom unit tests and checked for leaks and out-of-bounds access with Valgrind and cgdb.
- **CS61Classify in RISC-V assembly**: wrote assembly to classify handwritten digits with a simple ML algorithm, following the calling convention and using heap memory and file I/O; debugged with Venus, a calling-convention checker, and memcheck.

Publications & Preprints

- Chen, S. (2026). **Backtracking candidate elimination: a one-pass algorithm for the chip testing problem.** [arxiv]. An alternative $\mathcal{O}(n)$ time solution to the textbook divide-and-conquer version with fewer edge cases.
- Chen, S., Saban, N., Hargreaves, C., & Wang, H. (2026). **TreeAgent: a generalizable multi-agent framework for automated bias labeling in forestry via compiled expert rules and vision-language models.** ICML 2026 Workshop on AI for Science, [arxiv] [poster].

Activities & Leadership

- **VP of Research**, Society of Engineers, Scientists, and Entrepreneurs: collaborating with the Chen Institute on the AI4Science Symposium and mentoring 4 clubmates' CS research and class projects.

Honors & Awards

- **Innovation Catalysts**, Jacobs Institute for Design Innovation Feb 2026
- **Outstanding Project of the Year**, Chuying Innovation Talent Program (1 of 10 among 100+ projects) Feb 2025
- **Grand Slam**, Intl. Space Settlement Design Competition (ISSDC) Aug 2024
- **National Finalist**, China National College Mathematics Modeling Competition May 2024